PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

MAJOR PROJECT 1

(Topic selection)

A.Y-2025-2026

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Computer Science and Engineering
(Data Science)

AI-Powered Threat Intelligence System for Real-Time Cyber Threat Detection

Objectives

- To capture and analyze real-time network traffic and user logs for threat detection using Wireshark, Suricata, and Python-based tools like Scapy or PyShark.
- To detect known and unknown security threats using supervised models like Random Forest and SVM, and unsupervised models like Isolation Forest and K-Means.
- To build a real-time interactive dashboard using Streamlit, Plotly, and SQLite/Pandas to visualize threats, anomalies, and log trends effectively.
- To curate or simulate a balanced dataset of cyber threats using Python preprocessing,using customised dataset for attack simulation.
- To store network logs and model predictions using SQLite or MongoDB with Python libraries for efficient threat history analysis and pattern tracking.

Outcomes

- Real-time threat detection system using AI
- Multi-type attack classification with high accuracy
- Admin-friendly dashboard with alerts
- Scalable and modular framework for enterprise security

DiagnoX – Explainable AI for Personalized Healthcare via Multi-Modal Data Fusion

Objectives

- To build deep learning models that accurately diagnose by learning from diverse healthcare data sources like EHRs, imaging, and genomics using Pytorch/ TensorFlow.
- To use transformer-based NLP models to extract clinical insights from unstructured EHR text for patient profiling and early diagnosis using NLP with BERT / ClinicalBERT.
- To analyze genomic sequences using deep learning to identify mutations and risk factors for personalized treatment and disease prediction (CNN /Bi-LSTM models).
- To apply CNNs like ResNet and DenseNet for detecting pathological features in medical images to support AI-driven diagnostics.
- To integrate SHAP and LIME for generating transparent, interpretable explanations of AI-driven clinical predictions.

Outcomes

- Improved diagnostic accuracy by integrating EHRs, medical imaging, and genomics using deep learning.
- Personalized treatment recommendations enabled through patient-specific pattern analysis using GNNs and attention models.
- Robust diagnostic performance even with missing or incomplete data, using transformer-based intermediate fusion.
- Clinically interpretable AI outputs delivered through SHAP and LIME integration for transparent decision support.
- Standardized, scalable fusion framework adaptable across diverse healthcare datasets and populations.

Virtual Herbal Garden for AYUSH(Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy)

Objectives

- Implement efficient plant search and filtering using Hash Maps and Keyword Matching algorithms.
- Design and implement a 3D virtual garden environment in Unity that allows users to freely navigate and interact with medicinal plants using First-Person Controller and Collider-based Trigger Interactions.
- Enhance user learning with quizzes and flashcards using a custom Spaced Repetition algorithm.
- Detect fake herb input (wrong combinations) using Anomaly Detection techniques like Isolation Forest.

Outcomes

- Users can instantly search and filter medicinal plants by name, symptom, or category with high accuracy using optimized hash-based lookup and keyword matching.
- Users can explore and interact with medicinal plants in a 3D garden using first-person navigation and trigger-based actions.
- Users experience improved retention and recall of herbal knowledge through personalized quizzes and flashcards powered by a spaced repetition algorithm.
- The system automatically detects and flags suspicious or invalid herb combinations using Isolation Forest, ensuring safe and credible usage of medicinal information.

Thank You